The Briefing: something is coming

LEARNING TARGET · R1 I can describe what we know and don't know about 2026-XJ, and name the physics tools we'll need.

A · KEY VOCABULARY

Observation	Something we record or measure, like a position at a time.
Model	A useful "if": *if* the motion keeps going the same way, we can continue the pattern.
Prediction	An estimate of what comes next, made by continuing the pattern.
Testable question	One that measurements can answer. "Where will it be tomorrow?" — yes. "Who should pay for a mission?" — no; that also needs human decisions.

B · UNIT 1 ANCHOR + LANGUAGE OF MOTION

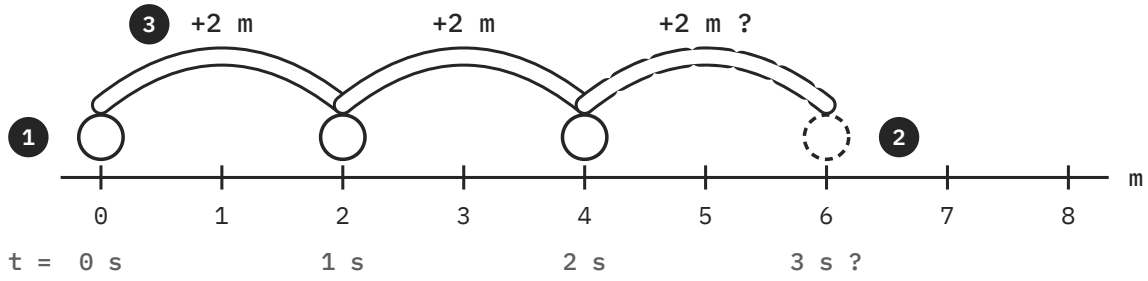
How fast, and where will it arrive?

- x position: where it is
- v velocity: how fast + which way
- a acceleration: how v changes

how fast · where · when · how big · made of · stop it?

C · COLOR IT · WATCH A PATTERN FORM

One dot marks the cart's position each second.



☐ **1 · BLUE** Observations: measured positions

☐ **2 · BLUE** Prediction: dashed outline

☐ **3 · PURPLE** The pattern: +2 m each second

D · WORKED EXAMPLE

A cart is at 2 m, 3 m, then 4 m, one second apart. Predict its position one second later.

GIVEN	EQUATION	WORK	ANSWER
x = 2, 3, 4 m 1 s between readings	next = last + change each second	change = 3 − 2 = +1 m next = 4 m + 1 m	**5 m** if the motion continues the same way

E · THE 2026-XJ THREAD

DAY 1 OF 22 · NEW TOOL: THE UNIT QUESTION

The desk has a sighting, not a forecast. Everything we learn in Unit 1 serves one question: how fast is 2026-XJ moving, and where will it arrive?

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else